Calculus Review II: Monotonicity, Integration Methods, Applications, and More

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§1.1 Derivative signs, extrema, and the mean value theorem

If f is differentiable on an interval I and

$$f'(x) > 0 \quad (x \in I),$$

then f is strictly increasing on I. If

$$f'(x) < 0 \quad (x \in I),$$

then f is strictly decreasing.

We follow the standard presentation of differentiation and integration from OpenStax, *Calculus*, Volumes 1 and 2, with the assumptions and error estimates made explicit below.[1] The proof uses the Lagrange mean value theorem. For $x_2 > x_1$,

$$f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1)$$

for some $\xi \in (x_1, x_2)$. If $f'(\xi) > 0$, then the difference is positive.

Suppose f is continuous near x_0 and differentiable in the punctured neighborhood. If the sign of f' changes from negative to positive at x_0, then $f(x_0)$ is a local minimum. If the sign changes from positive to negative, then $f(x_0)$ is a local maximum.

The compact criterion is:

$$(x - x_0)f'(x) > 0 \implies f(x_0) \text{ is a local minimum},$$
$$(x - x_0)f'(x) < 0 \implies f(x_0) \text{ is a local maximum}.$$

If

$$f'(x_0) = 0, \qquad f''(x_0) \neq 0,$$

then

$$f''(x_0) > 0 \implies x_0 \text{ is a local minimum},$$
$$f''(x_0) < 0 \implies x_0 \text{ is a local maximum}.$$

A useful mnemonic is that $f'' > 0$ looks like a smile and $f'' < 0$ looks like a frown.

If $f' \geq 0$ on an interval, the mean value theorem gives

$$f(y) - f(x) = f'(c)(y - x) \geq 0 \qquad (x < y).$$

So monotonicity is not a separate rule; it is a global consequence of local derivative sign.

Strict monotonicity needs care: $f' > 0$ everywhere is sufficient, not necessary. For $f(x) = x^3$, we have $f'(x) = 3x^2 \geq 0$ and $f'(0) = 0$, yet f is strictly increasing on all of $\mathbb{R}$. Similarly $f(x) = x^4$ has a strict local minimum at zero even though $f''(0) = 0$, so the second-derivative test is inconclusive in the zero case.

[1] OpenStax, *Calculus*, Vol. 1, Chapter 4, together with Vol. 2, Chapter 3.

§1.2 Concavity, asymptotes, and a global curve sketch

A curve segment is concave up if it lies above its tangent lines, and concave down if it lies below its tangent lines. In the usual calculus language,

$$f''(x) > 0$$

indicates concave up, while

$$f''(x) < 0$$

indicates concave down.

An inflection point is a point where the concavity changes. A necessary condition, when f'' exists, is

$$f''(x_0) = 0.$$

This condition alone is insufficient: vanishing or nonexistence of $f''(x_0)$ merely identifies a possible inflection point. The actual criterion is a change of concavity across the point.

An inflection point may have a horizontal or non-horizontal tangent, and in some examples the tangent is vertical or absent. What all cases share is a reversal of concavity, independent of the tangent's existence.

The asymptotic geometry of a graph has several distinct forms.

A vertical asymptote occurs when

$$\lim_{x \to x_0} f(x) = \pm\infty.$$

Then $x = x_0$ is a vertical asymptote.

A horizontal asymptote occurs when

$$\lim_{x \to +\infty} f(x) = c \quad \text{or} \quad \lim_{x \to -\infty} f(x) = c.$$

Then $y = c$ is a horizontal asymptote.

For an oblique asymptote

$$y = ax + b,$$

we require

$$\lim_{x \to \infty} [f(x) - (ax + b)] = 0.$$

The page derives

$$a = \lim_{x \to \infty} \frac{f(x)}{x},$$

and then

$$b = \lim_{x \to \infty} (f(x) - ax).$$

Both formulas follow from the defining limit after isolating the slope and intercept.

A systematic curve study proceeds through:

(i) determine the domain;

(ii) check symmetry and periodicity;

(iii) compute $f'(x)$ and $f''(x)$;

(iv) find critical points and possible inflection points;

(v) determine intervals of monotonicity and concavity;

(vi) find asymptotes;

(vii) locate intercepts or special points if useful;

(viii) draw the graph according to this information.

This is not a mechanical ritual. The derivative table tells how the curve moves; the second derivative table tells how it bends; asymptotes determine the large-scale shape.

Convexity can be read geometrically: the graph lies below chords and above supporting tangent lines. If f is differentiable and convex, then

$$f(y) \geq f(x) + f'(x)(y - x).$$

For $f(x) = x^2$, the tangent-line inequality reads $y^2 \geq x^2 + 2x(y - x)$, equivalently $(y - x)^2 \geq 0$. Thus the supporting-line property is a concrete algebraic consequence of convexity rather than a geometric slogan.

§1.3 Integration techniques as reverse differentiation

The integration section records standard formulas:

$$\int 0\,dx = C, \qquad \int x^{\mu}\,dx = \frac{x^{\mu+1}}{\mu+1} + C \quad (\mu \neq -1),$$

$$\int \frac{1}{x}\,dx = \ln|x| + C, \qquad \int a^x\,dx = \frac{a^x}{\ln a} + C.$$

Trigonometric formulas include

$$\int \sin x\,dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C,$$

$$\int \sec^2 x\,dx = anx + C, \qquad \int \csc^2 x\,dx = -\cot x + C,$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C, \qquad \int \frac{dx}{1+x^2} = \arctan x + C.$$

These antiderivative identities are valid only on intervals where their integrands and the chosen logarithm or inverse-trigonometric branches are defined; in particular the exponential formula requires $a > 0$, $a \neq 1$. The page also lists logarithmic forms such as

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln|x + \sqrt{x^2+a^2}| + C.$$

For a composite function, the substitution rule is

$$\int f(\varphi(v))\varphi'(v)\,dv = \int f(u)\,du, \qquad u = \varphi(v).$$

Equivalently,

$$\int f(u)\,du = \int f(\varphi(v))\varphi'(v)\,dv.$$

One direction recognizes a derivative already present, while the other introduces a variable that simplifies the integrand. The first recognizes a derivative already present; the second introduces a new variable to simplify the integral.

From

$$d(uv) = u\,dv + v\,du,$$

one obtains

$$\int u\,dv = uv - \int v\,du.$$

Integration by parts complements substitution. Its use depends on choosing u so that differentiating it simplifies the expression and choosing dv so that it can be integrated.

A rational function is

$$\frac{P(x)}{Q(x)},$$

where P, Q are polynomials and $Q \neq 0$. If $\deg P \geq \deg Q$, first divide polynomials to write it as a polynomial plus a proper rational function.

The pages list the elementary partial fraction forms. A real linear factor $(x-a)^k$ contributes

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_k}{(x-a)^k}.$$

An irreducible quadratic factor $(x^2+px+q)^k$ with $p^2 - 4q < 0$ contributes

$$\frac{M_1 x + N_1}{x^2+px+q} + \cdots + \frac{M_k x + N_k}{(x^2+px+q)^k}.$$

Thus integration of rational functions reduces to two basic integral types:

$$\int \frac{dx}{(x-a)^k}, \qquad \int \frac{Mx+N}{(x^2+px+q)^k}\,dx.$$

For the second type, split the numerator as a derivative part plus a remainder:

$$Mx + N = \frac{M}{2}(2x+p) + \left(N - \frac{M}{2}p\right).$$

Then

$$\int \frac{Mx+N}{(x^2+px+q)^k}\,dx = \frac{M}{2} I_k^1 + \left(N - \frac{M}{2}p\right) I_k^2,$$

where

$$I_k^1 = \int \frac{d(x^2+px+q)}{(x^2+px+q)^k},$$

and

$$I_k^2 = \int \frac{dx}{(x^2+px+q)^k}.$$

Complete the square by setting

$$t = x + \frac{p}{2}, \qquad a^2 = q - \frac{p^2}{4} > 0.$$

Then

$$x^2 + px + q = t^2 + a^2.$$

For $k = 1$,

$$I_1^2 = \frac{1}{a}\arctan\frac{t}{a} + C.$$

For higher k, the recurrence shown on the page is

$$I_{k+1}^2 = \frac{1}{2ka^2}\left(\frac{t}{(t^2+a^2)^k} + (2k-1)I_k^2\right).$$

The notes next handle rational functions of $\sin x$ and $\cos x$. Use the tangent half-angle substitution

$$t = an\frac{x}{2}, \qquad -\pi < x < \pi.$$

Then

$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad anx = \frac{2t}{1-t^2},$$

and

$$dx = \frac{2\,dt}{1+t^2}.$$

Thus

$$\int R(\sin x, \cos x)\,dx = \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right)\frac{2\,dt}{1+t^2},$$

which becomes a rational integral in t.

§1.4 Definite Integrals and Averages

The page defines the definite integral. Divide $[a, b]$ by

$$a = x_0 < x_1 < \cdots < x_n = b,$$

let

$$\Delta x_i = x_i - x_{i-1},$$

and choose sample points $\xi_i \in [x_{i-1}, x_i]$. The Riemann sum is

$$\sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

If the limit exists as the mesh

$$\lambda = \max_i \Delta x_i$$

tends to zero, then

$$\int_a^b f(x)\,dx$$

is the definite integral.

The pages list sufficient conditions for integrability: continuous functions, monotone bounded functions, and bounded functions with finitely many discontinuities are integrable on closed intervals.

The standard properties are:

$$\int_a^b [\alpha f(x) + \beta g(x)]\,dx = \alpha\int_a^b f(x)\,dx + \beta\int_a^b g(x)\,dx,$$

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx, \qquad \int_a^a f(x)\,dx = 0,$$

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

If $f(x) \geq 0$ on $[a, b]$, then

$$\int_a^b f(x)\,dx \geq 0.$$

If $f \le g$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

Theorem 1.4.1 (Integral mean value theorem)
If f is continuous on $[a,b]$, then there exists $c \in [a,b]$ such that

$$\int_a^b f(x)\,dx = f(c)(b-a).$$

If f is continuous on $[a,b]$, then there exists $\xi \in [a,b]$ such that

$$\int_a^b f(x)\,dx = f(\xi)(b-a).$$

If g does not change sign, the weighted form is

$$\int_a^b f(x)g(x)\,dx = f(\xi)\int_a^b g(x)\,dx.$$

For a continuous f, let $m = \min_{[a,b]} f$, $M = \max_{[a,b]} f$. Then $m(b-a) \le \int_a^b f \le M(b-a)$ when $a < b$. Dividing by $b-a$ places the average inside $[m,M]$, and the intermediate value theorem supplies $f(c)$ equal to that average. This proves the integral mean value theorem; the weighted variant also needs g integrable with constant sign.

Area between curves is computed by

$$A = \int_a^b [f(x) - g(x)]\,dx$$

when $f \ge g$. If horizontal slicing is better,

$$A = \int_c^d [x_{right}(y) - x_{left}(y)]\,dy.$$

The volume of a solid of revolution around the x-axis is

$$V = \pi \int_a^b [f(x)]^2\,dx.$$

The shell method has the form

$$V = 2\pi \int_a^b x f(x)\,dx$$

when rotating a region around the y-axis.

Arc length of $y = f(x)$ is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx.$$

For a parametric curve $x = x(t), y = y(t)$,

$$L = \int_\alpha^\beta \sqrt{(x'(t))^2 + (y'(t))^2}\,dt.$$

§1.5 Improper integration and quadrature error

The last pages include reminders about improper integrals. If the interval is unbounded,

$$\int_a^{+\infty} f(x)\,dx = \lim_{R\to+\infty} \int_a^R f(x)\,dx.$$

If the integrand has a singularity at b, then

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx.$$

Convergence depends on the existence of these limits.

A standard comparison is the p-integral:

$$\int_1^\infty \frac{dx}{x^p}$$

converges iff $p > 1$, while

$$\int_0^1 \frac{dx}{x^p}$$

converges iff $p < 1$.

For $f \in C^2([a,b])$, the one-panel midpoint and trapezoid approximations satisfy the error bounds

$$\left| \int_a^b f(x)\,dx - (b-a) f\left(\frac{a+b}{2}\right) \right| \le \frac{(b-a)^3}{24} \|f''\|_\infty,$$

$$\left| \int_a^b f(x)\,dx - \frac{b-a}{2}(f(a)+f(b)) \right| \le \frac{(b-a)^3}{12} \|f''\|_\infty.$$

For $f(x) = x^2$ on $[0,1]$, the exact integral is $1/3$; the midpoint estimate $1/4$ errs by $1/12$, and the trapezoidal estimate $1/2$ errs by $1/6$. The sign of each error is consistent with the graph's convexity. These computations expose the actual role of the second derivative in quadrature.